Circles and Parabolas

ANSWER KEY

Circles

1 Write the standard-form equation of the circle with center $(3, -2)$ and radius 5.

$(x-3)^2+(y+2)^2=25$.

2 Consider the circle $x^2+y^2-6x+4y-12=0$.

a Write the equation in standard form by completing the square. $(x-3)^2+(y+2)^2=25$.

b State the center and radius. Center $(3, -2)$, radius 5.

3 Determine whether the point $(6, 2)$ lies inside, on, or outside the circle $(x-3)^2+(y+2)^2=25$.

$(6-3)^2+(2+2)^2=9+16=25$, so the point lies on the circle.

Parabolas with vertex at the origin

4 For the parabola $y^2=8x$, find:

a the focus $(2, 0)$

b the directrix $x=-2$

c the direction of opening opens right

5 A parabola has vertex at the origin, opens upward, and passes through the point $(4, 2)$.

a Find its equation in the form $x^2=4py$. $16=4p(2)$ gives $p=2$, so $x^2=8y$.

b State the coordinates of the focus. $(0, 2)$.

6 A satellite dish has a parabolic cross-section 2 m wide and 0.5 m deep. Taking the vertex at the origin opening upward, find the distance from the vertex to the focus (where the receiver is placed).

The edge point $(1, 0.5)$ lies on $x^2=4py$: $1=4p(0.5)=2p$, so $p=0.5$ m. The receiver sits 0.5 m above the vertex.